

WILLIAM EWANCHUK

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EDUCATION

University of Alberta | B.Sc. Computer Engineering Co-op | Expected May 2027

EXPERIENCE

Full Stack Developer/DevOps

April 2026 – Current

Company: North American Construction Group

Stack - Azure DevOps, Static Web Apps, Azure Container Apps/KEDA, Docker/ACR, .NET 10, .NET Core, React/Tanstack, React Native, SQLite, OpenTelemetry, Cloudflare

- Contributed to the deployment and rewrite of **WorX**, an internal and subcontractor-facing application for coordinating mining equipment availability, parts sourcing, technician scheduling, and work order creation and completion.
- Containerized the full application with **Docker**, using dependency and NuGet caching stages to cut build times by **80%**; configured local Azurite, MSSQL, and Table Storage emulators and published production images to ACR.
- Owned the **Azure Blob Storage** implementation from design through production, using Entra-authenticated user-delegation SAS tokens for secure image, video, and PDF attachments and an optimistic pre-submission upload flow.
- Instrumented monitoring with **OpenTelemetry** and Application Insights, enabling per-user log-level controls and request tracing to debug issues encountered by beta testers in production.
- Designed the beta release pipeline end-to-end, including pending migration detection, required changelog validation, ACR image publishing, and blue/green **Azure Container Apps** revision deployments for staging and production.
- Built offline mobile support using **optimistic replicated state**, allowing dependent operations to proceed locally and synchronize in order on reconnect.

Software Engineering Intern

November 2024 – August 2025 (10 months)

Company: Stepscale

Stack - Django, PostgreSQL, LangGraph, LangChain, OpenRouter, AWS Elastic Beanstalk, CloudWatch, Docker, Pipedream, Stripe, Amplitude, Redis, GitHub Actions

- Built an isolated Docker sandbox for LangGraph agents to run Python file generation and analysis with automated output validation.
- Integrated Amplitude analytics to track tool call errors, navigation, and landing page bounce rates, identifying failing flows and high drop-off pages.
- Designed and secured backend endpoints for **Stripe** billing, including pricing tiers, subscriptions, and webhooks.
- Implemented a **Redis** caching layer to pre-warm LangChain conversation chains for connected integrations, reducing redundant inference requests and improving response latency by **40%**.
- Monitored and debugged deployments with AWS CloudWatch and GitHub Actions across dev and prod environments.

PROJECTS

LittleBrick 3D Printing – E-Commerce Platform

GitHub | Live Demo

Stack - Spring Boot, PostgreSQL, Next.js, TypeScript, Stripe, Shippo, Google Maps, Resend, Docker, Nginx, Terraform, Ansible, Tailscale, Cloudflare Tunnels, Actions Runners

- Built a **3D printing e-commerce platform** for requesting custom prints, calculating live shipping rates, handling payments, and tracking orders through delivery for a small local business.
- Estimated live parcel shipping rates/quotes from cart item dimensions/weights algorithmically before checkout.
- Integrated Google Maps to display live shipment and destination locations, geocoding addresses server-side.
- Built an interactive 3D product preview, detecting regions 3mf/stl to render live color preview.
- Coordinated Stripe and Shippo webhooks to drive the order lifecycle, with Stripe events advancing order status and Shippo events updating shipment tracking.
- Replaced Spring Security's default session handling with PostgreSQL-backed bearer sessions, enabling RBAC for administrative endpoints.
- Provisioned the production server on Proxmox with Terraform and Ansible, exposing each environment through reverse-proxied Cloudflare Tunnels.

EKS Traffic Driven Scaling Platform: DevOps/SRE

GitHub | Live Demo

Stack - Kubernetes, Helm, EKS, KEDA, Grafana, Prometheus, Terraform, Python, Docker, Redis, AWS CLI, GitLab

- Deployed a FastAPI service to AWS EKS using Terraform, with automated TLS, monitoring, and PromQL autoscaling to benchmark Gunicorn worker and thread configurations
- Automated a CI/CD style deployment pipeline for AWS EKS using AWS CLI and Helm, including Route 53 DNS updates and NGINX Ingress TLS certificate provisioning with Let's Encrypt
- Enabled traffic based autoscaling with KEDA and Prometheus metrics, monitored in Grafana and load tested with k6, sustaining **5k+ RPS at 500 VUs**
- Built a lightweight dashboard for internal pod metrics and optimized CPU metric collection with a Redis shared cache to avoid redundant per container polling

SKILLS

Languages - Java, C#, Python, TypeScript, Bash, SQL, C, C++

Frameworks - Spring Boot, .NET, React/React Native, Django, FastAPI

Databases - PostgreSQL, MySQL, SQLite, MongoDB, Redis

Cloud and DevOps - Kubernetes, KEDA, Docker, Azure (Container Apps/ACR/Static Web Apps/Blob Storage), AWS (EKS/EC2/S3/RDS/CloudWatch), Cloudflare, Tailscale, Nginx, Terraform, Ansible, Helm

Dev Tools and Observability - Git, GitHub Actions, GitLab CI, Azure DevOps, Grafana, Prometheus, OpenTelemetry, Application Insights, CloudWatch, JUnit